

Jacobians Part 2

EGR 445/545

GVSU

Plan for Today

- Jacobian review
- Two link Jacobian
- Three link with twist
- Three link planar

Conceptual Review

- what is a Jacobian?
- what is a singularity?
 - mathematically
 - physically

How to Find the Jacobian

- find $\bar{v}_{tip}$ and $\bar{\omega}_N$ symbolically using `sympy` and velocity propagation
- take partial derivatives to find $\mathbf{J}$
- find singularity conditions based on $\|\mathbf{J}\| = 0$

Velocity Propagation

Conceptually:

- we work from the base out to the tip
- velocity of the end of a link is velocity at the base of the link plus the rotational part:
 - $\bar{v}_{tip} = \bar{v}_{base} + \bar{\omega} \times \bar{r}$
 - $\bar{\omega}$ is the total rotational velocity of the link, including rotation of all lower numbered joints
 - $\bar{r}$ is the position vector of the tip with respect to the base of the link
 - vectors for cross products must be in the same frame

Velocity Propagation Equations

- eqn. 5.46: ${}^i v_{i+1} = {}^i v_i + {}^i \bar{\omega}_i \times {}^i P_{i+1}$
 - example: ${}^1 v_2 = {}^1 v_1 + {}^1 \bar{\omega}_1 \times {}^1 P_2$
- eqn. 5.47K: ${}^{i+1} v_{i+1} = {}_i^{i+1} \mathbf{R} \cdot {}^i v_{i+1}$
 - example: ${}^2 v_2 = {}_1^2 \mathbf{R} ({}^1 v_1 + {}^1 \bar{\omega}_1 \times {}^1 P_2) = {}_1^2 \mathbf{R} {}^1 v_2$
- eqn. 5.45: ${}^{i+1} \bar{\omega}_{i+1} = {}_i^{i+1} \mathbf{R} {}^i \bar{\omega}_i + \dot{\theta}_{i+1} {}^{i+1} \hat{Z}_{i+1}$

Two-Link Planar Jacobian

- two link with arbitrary angles
- how do we find the singularity condition(s)?
- walk through all the steps using `sympy`

`two_link_Jacobian_full_example.ipynb`

Three Link with Twist

- add third link to two-link planar, but $\alpha_2 = 90^\circ$

`three_link_with_twist_Jacobian_example.ipynb`

Jacobian Learning Activity

- LA 12